

Cover Letter

I am particularly interested in this position because it focuses on developing self-learning artificial intelligence systems and autonomous experimentation frameworks for dynamic healthcare environments. My research background in computational biology and machine learning has involved building predictive models and scalable computational pipelines to analyze complex biological datasets and deploy reproducible analytical tools for biomedical research. During my Ph.D. at IIIT-Delhi, I developed machine learning models for protein function prediction and immunological data analysis, integrating heterogeneous biological datasets and implementing deployable computational workflows. More recently, I have worked with representation-learning approaches using protein language models, such as ProtBERT and ESM2, to extract functional signals from biological sequences. My experience with Python-based machine learning frameworks, high-performance computing environments, and reproducible research pipelines aligns well with the development of reliable AI systems and data-driven workflows that can support biomedical discovery and digital health applications.